Lecture 14?

Tuesday, 17 October 2023 10:37 AM

Solid state NMR (very different from liquid state)

ESR
↳ electron spin resonance
very sensitive but only applicable to certain applications

+ Low frequency & energy
+ Exciting spin states of nuclei (& e^-) } least amount of energy required but need sensitive detectors.
↓
splitting degenerate spin states
(need to actually spin the sample

Larmor frequency

frequency of precession of ^1H ; $\omega = -\gamma B$

ω → frequency
γ → gyromagnetic ratio $\propto \frac{1}{\text{mass of nucleus}}$

$$\gamma = \frac{-eg}{2m}$$

I → nuclear spin quantum no.

NMR Receptivity